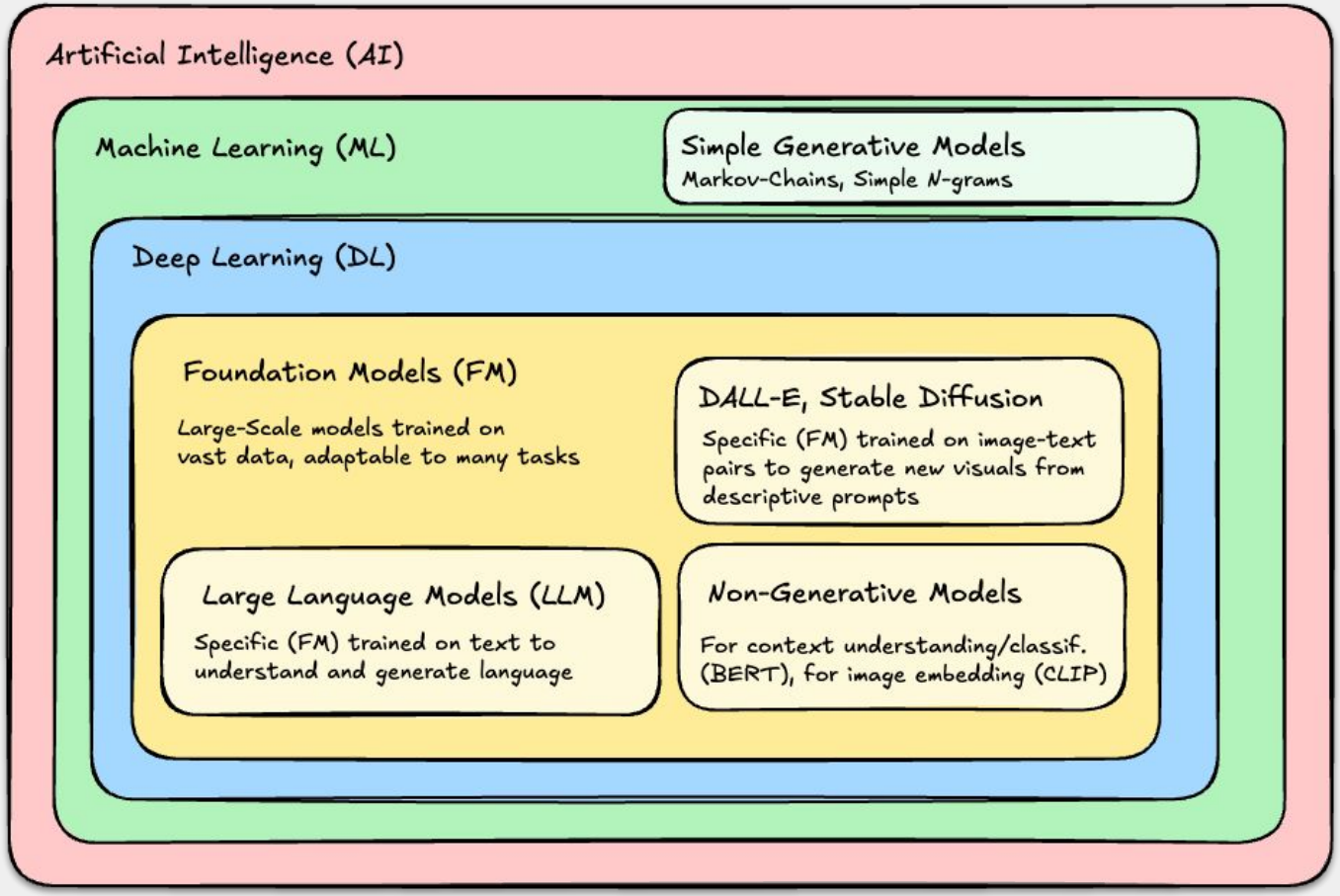
PPGEEC2327

Deep Dive Into LLM

Lecture 2

Ivanovitch Silva

March, 2026



How it's built (methodology) → Foundation Models (FM)

What it does (capability) → Generative AI

ImageNet Classification with Deep Convolutional Neural Networks

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Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

1 Introduction

Current approaches to object recognition make essential use of machine learning methods. To improve their performance, we can collect larger datasets, learn more powerful models, and use better techniques for preventing overfitting. Until recently, datasets of labeled images were relatively small — on the order of tens of thousands of images (e.g., NORB [16], Caltech-101/256 [8, 9], and CIFAR-10/100 [12]). Simple recognition tasks can be solved quite well with datasets of this size, especially if they are augmented with label-preserving transformations. For example, the current-best error rate on the MNIST digit-recognition task (<0.3%) approaches human performance [4]. But objects in realistic settings exhibit considerable variability, so to learn to recognize them it is necessary to use much larger training sets. And indeed, the shortcomings of small image datasets have been widely recognized (e.g., Pinto et al. [21]), but it has only recently become possible to collect labeled datasets with millions of images. The new larger datasets include LabelMe [23], which consists of hundreds of thousands of fully-segmented images, and ImageNet [6], which consists of over 15 million labeled high-resolution images in over 22,000 categories.

To learn about thousands of objects from millions of images, we need a model with a large learning capacity. However, the immense complexity of the object recognition task means that this problem cannot be specified even by a dataset as large as ImageNet, so our model should also have lots of prior knowledge to compensate for all the data we don't have. Convolutional neural networks (CNNs) constitute one such class of models [16, 11, 13, 18, 15, 22, 26]. Their capacity can be controlled by varying their depth and breadth, and they also make strong and mostly correct assumptions about the nature of images (namely, stationarity of statistics and locality of pixel dependencies). Thus, compared to standard feedforward neural networks with similarly-sized layers, CNNs have much fewer connections and parameters and so they are easier to train, while their theoretically-best performance is likely to be only slightly worse.

Learning Transferable Visual Models From Natural Language Supervision

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arXiv:2103.00020v1 [cs.CV] 26 Feb 2021

Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at https://github.com/OpenAI/CLIP.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019). Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of "text-to-text" as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image captions learn useful image representations. They converted the title, description, and hashtag metadata of images in the YFCC100M dataset (Thomee et al., 2016) into a bag-of-words multi-label classification task and showed that pre-training AlexNet (Krizhevsky et al., 2012) to predict these labels learned representations which preformed similarly to ImageNet-based pre-training on transfer tasks. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image

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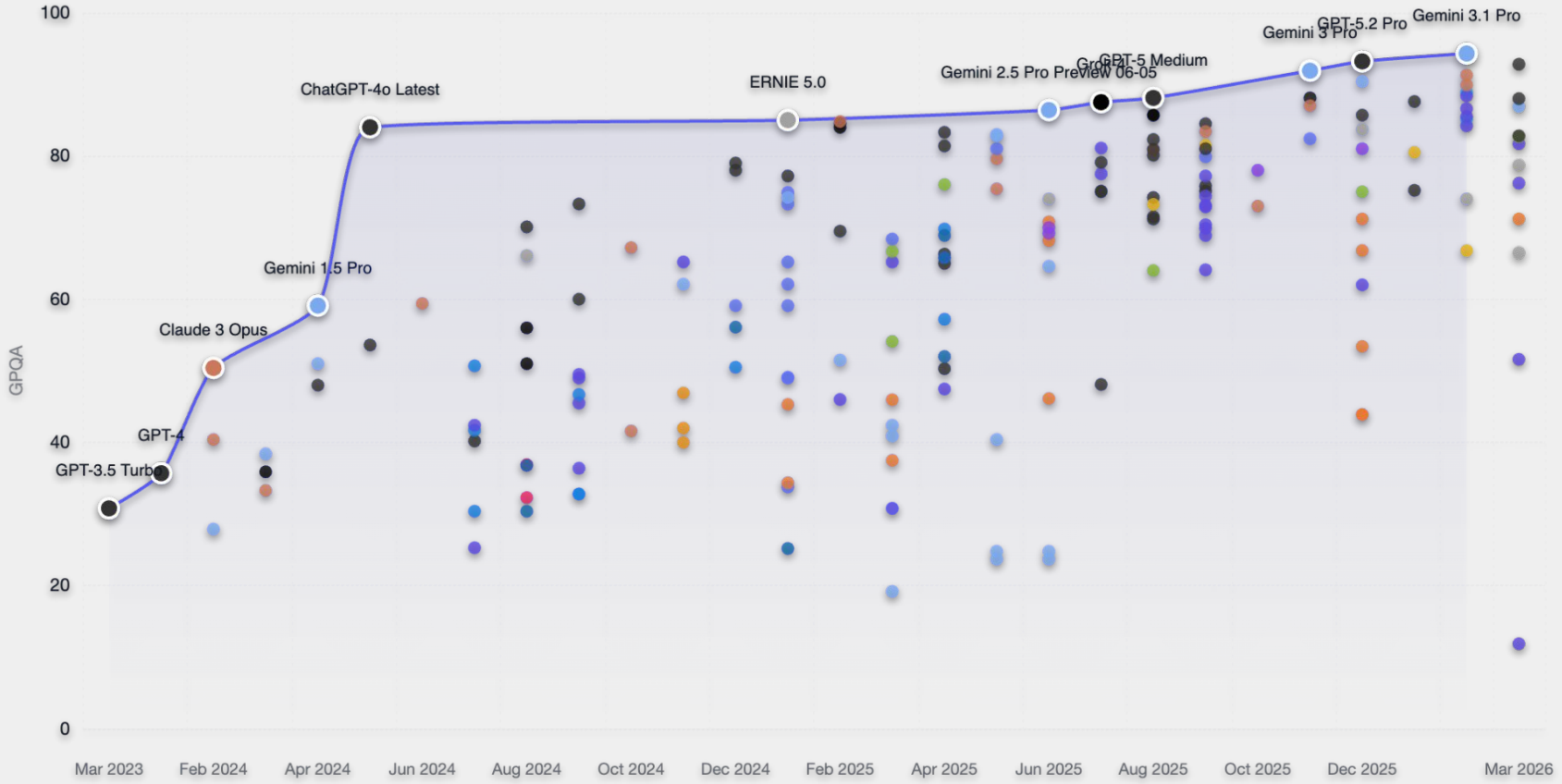
Graduate-Level Google-Proof Q&A (GPQA). Are LLMs Becoming Commodities?



https://llm-stats.com

Last update: 26 march 2026

Performance evaluation of a compression algorithm for wireless sensor networks in monitoring applications

Publisher: IEEE

Cite This
PDF

Ivanovitch M. D. Silva ; Luiz Affonso Guedes ; Francisco Vasques All Authors

19 Cites in Papers
250 Full Text Views

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Abstract

Document Sections

1. Introduction
2. Wireless Sensor Network
3. Compression Algorithm

Abstract:
Wireless sensor network (WSN) is an emerging technology that targets multiple applications in the different environments. Its infrastructure is composed of a large number of sensor nodes with a limited physical capacity and low cost. The energy consumption must be as optimized as possible in order to extend its lifetime. The use of data compression techniques can be an advantage in the WSN context, once these techniques eliminate transmission of redundant information and consequently can be adopted to minimize the consumption of energy in the sensor nodes. WSN for monitoring applications can benefit from this technique as it may maximize the lifetime of batteries. The main motivation of this paper is to investigate the

More Like This

ESMAC: Enhanced S-MAC Protocol on Network Topologies to Improve Energy-Efficient in Wireless Sensor Networks

2023 10th International Conference on Electrical and Electronics Engineering (ICEEE)

Published: 2023

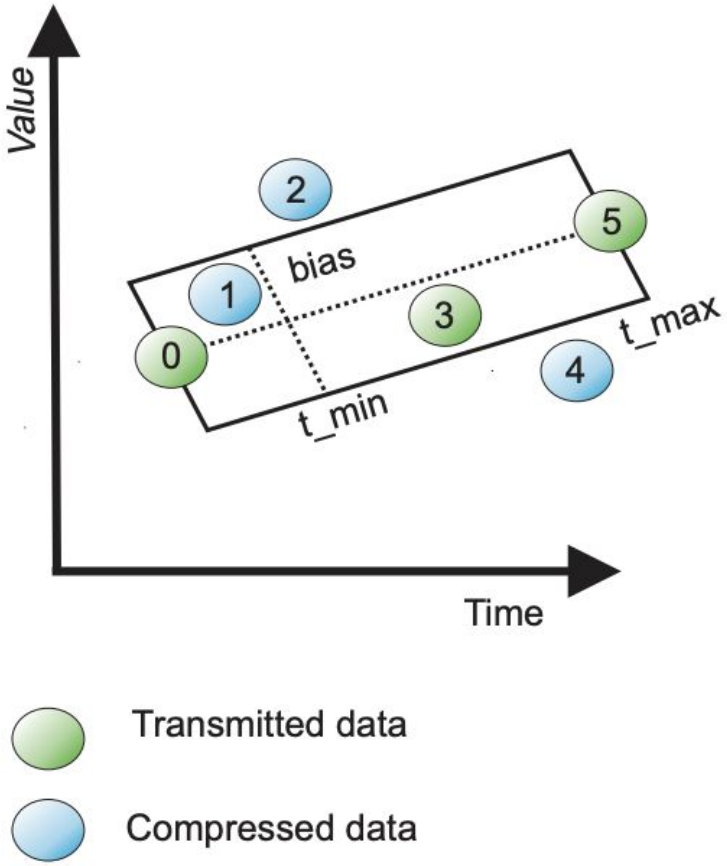


Figure 3. Swinging Door example.

Prompt:

Estou revisando um artigo e gostaria de propor uma nova legenda para a Figura 3.

ChatGPT

Aqui vai uma proposta de legenda mais clara, técnica e alinhada ao conteúdo da figura e do texto do artigo:

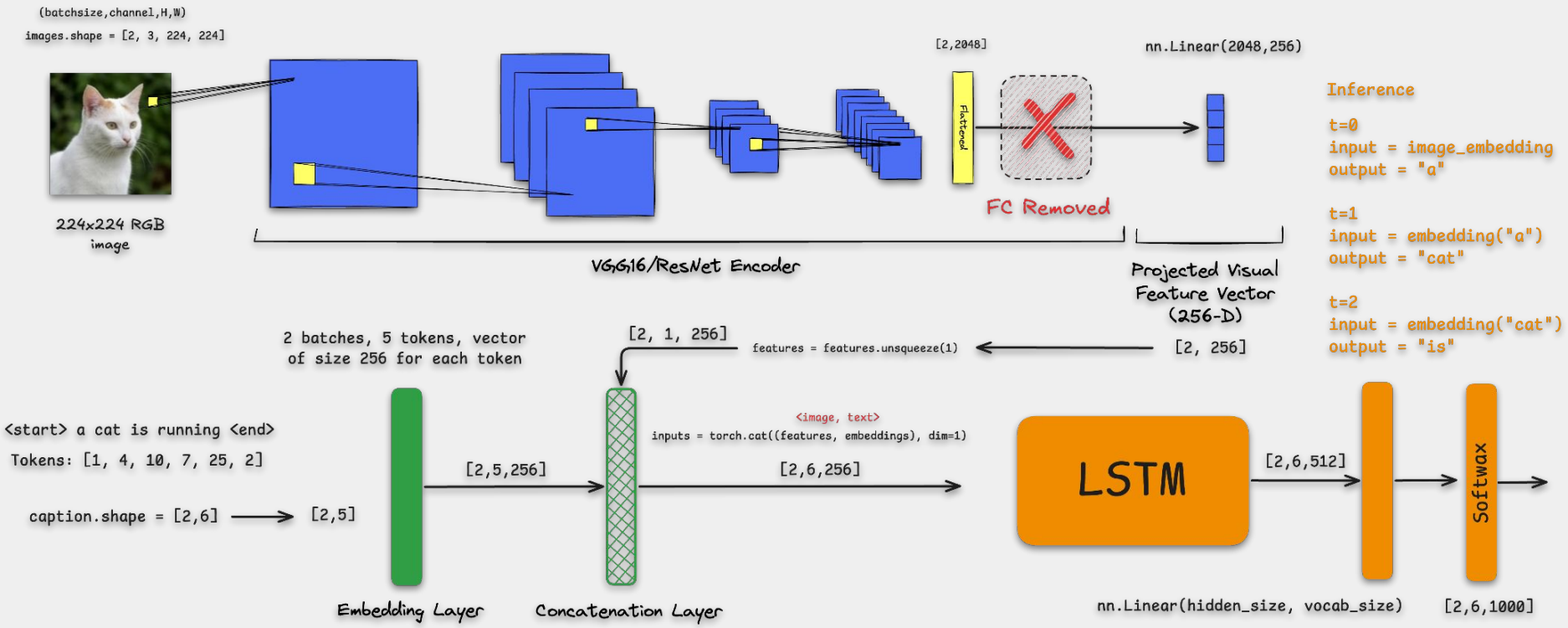
Proposta de legenda (versão revisada):

> **Figure 3. Illustration of the Swinging Door temporal compression mechanism.** The sliding time window defined by *t_min*, *t_max*, and *bias* determines whether incoming samples are transmitted or suppressed. Green points represent transmitted data, while blue points indicate compressed (discarded) samples. The example highlights how the algorithm preserves relevant deviations while reducing redundant transmissions. Performance_evaluation_of_a_com...

Se quiser uma versão um pouco mais "assertiva" (estilo IEEE mais direto):

> **Figure 3. Swinging Door temporal compression example.** The time window defined by *t_min*, *t_max*, and *bias* filters incoming samples, transmitting only relevant data (green) and discarding redundant points (blue).

CNN-LSTM Architecture for Image Captioning



Learning

Knowing

Information is directly acquired from reliable sources such as books, teachers, lectures, and courses.

Focus on absorbing and retaining information.

Discovering

Self-directed learning that happens when one ventures out to find new information

Driven by curiosity and inherently exploratory.

Learning that takes place when information or experiences are exchanged between individuals or groups.

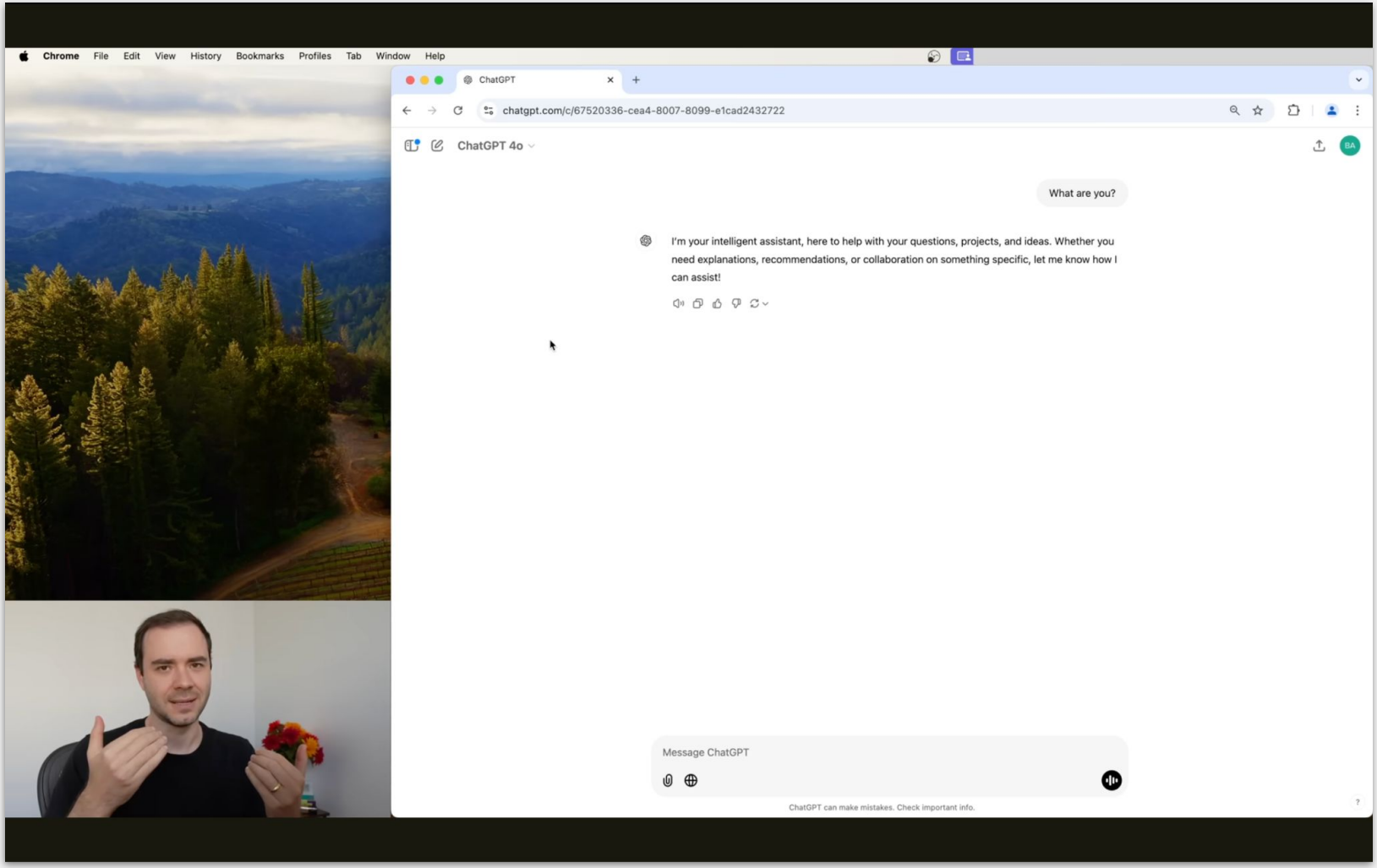
Teaching others reinforces one's own understanding.

Sharing

'Learning by doing', based on the premise that direct experience is one of the most potent forms of learning.

Involves real-life situactions

Experiencing



https://www.youtube.com/watch?v=7xTGNNLPyMI&list=PLAqhIrjkxbuW9U8-vZ_s_cjKPT_FqRStI

Assignment 1

LLM Deep Dive (Reverse Classroom)

Overview

In this assignment, you will study Large Language Models (LLMs) through a **reverse classroom approach**.

Instead of attending a lecture on the topic, you will:

- Study a comprehensive Deep Dive lecture (by Andrej Karpathy)
- Reconstruct the concepts in your own way
- Teach them as if you were the instructor

The goal **is not to reproduce the content**, but to demonstrate that you can understand, structure, and communicate complex ideas independently.

This assignment also includes a **peer interaction component,** designed to promote reflection and intellectual exchange in a natural and unscripted way.

Learning Objectives

By completing this assignment, you should be able to:

- Explain the end-to-end pipeline of a modern LLM
- Describe the role of tokenization, embeddings, and Transformers
- Understand the training paradigm (next-token prediction)
- Identify limitations and open challenges in LLMs
- Communicate technical ideas clearly at a graduate level
- Critically engage with another person’s explanation

Part I - Your Lecture (Individual Work)

You will create your own explanation of the Deep Dive content.

Format

- Platform: Loom (www.loom.com)
- Maximum: 10 minutes per video
- Expected: 3 videos (≈ 25–30 minutes total)

Suggested Structure (flexible, not rigid)

You are encouraged, but not forced, to organize your videos around:

1. The Big Picture

Explain the overall pipeline:

- Tokens → Embeddings → Transformer → Output
- What the model is actually doing
- Focus on clarity and conceptual flow.

2. Technical Deep Dive

Explain core technical mechanisms in more detail:

- Attention mechanism (intuition + formal perspective)
- Transformer blocks (self-attention, feed-forward layers)
- Training process (objective, data, scaling)
- Inference behavior
- Students are encouraged to include diagrams, equations, or examples where appropriate

3. Critical Reflection

Provide a critical and analytical perspective:

- What are the limitations of LLMs?
- What aspects were unclear or under explained in the source material?
- What are common misconceptions?
- What questions remain open?
- This section should demonstrate independent thinking, not just reproduction of content.

Part II - Peer Interaction (organic, asynchronous)

You will watch the videos produced by your partner (from the first class activity).

Your role is simple:

- Engage with your colleague’s explanation in a thoughtful and honest way.

Part II - How to do it

Use the comment feature in Loom.

There is no rigid template, but your interaction should naturally include:

- What you found clear
- What you did not fully understand
- What you think is missing
- At least one meaningful question

After that:

- The author is expected to respond
- A short back-and-forth may emerge

Important Principles

- This is not a checklist exercise
- This is not scripted interaction
- Do not coordinate responses in advance
- Do not treat this as a formality

The goal is to create a genuine technical exchange

Academic Integrity

You may use AI tools to:

- organize your ideas
- improve clarity

However:

- The explanation must reflect your own understanding
- You must be able to defend everything you present
- Blind copying or passive generation is not acceptable

Grading Rubric

A - Excelente

- Demonstrates clear and solid understanding of LLM concepts
- Explanation is well-structured and easy to follow
- Shows depth (goes beyond surface-level description)
- Includes critical thinking (questions, limitations, insights)
- Peer interaction is genuine, thoughtful, and technically meaningful

B - Good

- Good understanding, with minor gaps
- Explanation is mostly clear
- Some depth, but not consistent
- Some critical reflection, but limited
- Peer interaction is relevant, but not very deep

C — Satisfactory

- Basic understanding of concepts
- Explanation is somewhat superficial or uneven
- Limited depth
- Little critical thinking
- Peer interaction exists, but is generic or shallow

19/03 Week 01
Overview & Introduction

26/03 Week 02
Fundamentals - Deep Dive to LLM

09/04 Week 03
Fundamentals - Why attention

16/04 Week 04
Fundamentals - Transformer

23/04 Week 05
Fundamentals - How to implement the components

30/04 Week 06
Nano-Project

07/04 Week 07
Prompting for Effective LLM Reasoning and Planning

14/07 Week 08
Agentic Workflow

21/05 Week 09
Building Agents

28/05 Week 10
Multi-Agent System

11/06 Week 11
Project

18/06 Week 12
Project

25/06 Week 13
Project <meetup>

02/07 Week 14
Project <presentation>

09/07 Week 15
Project <presentation>

Evaluation Criteria

There will be no exams. Instead, students will complete weekly assignments

[A] Submitted all assignments & Fully met the requirements

[B] Failed to submit at least 1 assignment & Partially met the requirements of at least 2 assignments

[C] Failed to submit at least 2 assignments & Partially met the requirements of at least 3 assignments

[D] Failed to submit at least 3 assignments & Partially met the requirements of at least 4 assignments